

Distance vs Displacement

- **Distance (d)**
 - Total length of the path travelled
 - Measured in meters
 - scalar
- **Displacement ($\vec{d}$)**
 - Change in position (Δx) regardless of path
 - $\Delta x = x_f - x_i$
 - Measured in meters
 - vector

Vectors and Scalars

- Vector quantities need both **magnitude** (size or numerical value) and **direction** to completely describe them
 - Will use + and - signs to indicate vector directions
- Scalar quantities are completely described by magnitude only

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Elapsed time}} \cdot \frac{x_2 - x_1}{t_2 - t_1} \quad \text{SI unit: } \frac{\text{m}}{\text{s}}$$

Graphs. On a graph of x versus t , v_{avg} is the **slope** of the straight line that connects two particular points on the $x(t)$ curve: one is the point that corresponds to x_2 and t_2 , and the other is the point that corresponds to x_1 and t_1 . Like displacement, v_{avg} has both magnitude and direction (it is another vector quantity). Its magnitude is the magnitude of the line's slope. A positive v_{avg} (and slope) tells us that the line slants upward to the right; a negative v_{avg} (and slope) tells us that the line slants downward to the right. The average velocity v_{avg} always has the same sign as the displacement Δx because t in Eq. is always positive.

$$\text{Average speed} : \frac{\text{Distance}}{\text{elapsed time}} = \frac{l}{\Delta t} \quad \text{unit: } \frac{\text{m}}{\text{s}}$$

Instantaneous Velocity and Speed

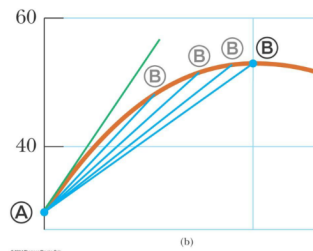
The **instantaneous velocity** indicates how fast the car moves and the direction of motion at each instant of time.

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}.$$

The **speed** is the magnitude of the instantaneous velocity

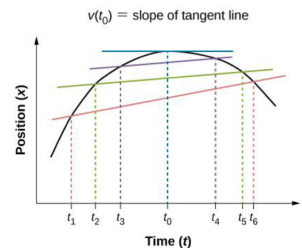
Instantaneous Velocity, graph

- The instantaneous velocity is the **slope of the line tangent** to the x vs. t curve
- The **blue lines** show that as Δt gets smaller, they approach the green line
- This would be the **green line**



Instantaneous Speed

- The instantaneous speed is the magnitude of the instantaneous velocity
- Remember that the average speed is **not** the magnitude of the average velocity.



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ch.2.3

Acceleration:

Acceleration

When a particle's velocity changes, the particle is said to undergo **acceleration** (or to accelerate). For motion along an axis, the **average acceleration** a_{avg} over a time interval Δt is

$$a_{\text{avg}} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}, \quad (2-7)$$

where the particle has velocity v_1 at time t_1 and then velocity v_2 at time t_2 . The **instantaneous acceleration** (or simply **acceleration**) is

$$a = \frac{dv}{dt}. \quad (2-8)$$

In words, the acceleration of a particle at any instant is the rate at which its velocity is changing at that instant. Graphically, the acceleration at any point is the slope of the curve of $v(t)$ at that point. We can combine Eq. 2-8 with Eq. 2-4 to write

$$a = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2x}{dt^2}. \quad (2-9)$$

In words, the acceleration of a particle at any instant is the second derivative of its position $x(t)$ with respect to time.

sensation: Body reacts when acceleration changes $\Rightarrow$ resulting in G force
G unit: Measures acceleration relative to Earth's gravity. $g = 9.8 \text{ m/s}^2$



If the signs of the velocity and acceleration of a particle are the same, the speed of the particle increases. If the signs are opposite, the speed decreases.